THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** material	An object is made of material.
2	**(a)** sugar	Sugar dissolves.
3	**(a)** copper	Metals conduct heat well.
4	**(b)** transparent	We can see clearly through them.
5	**(a)** sand	Sand does not dissolve.
6	**(c)** poor conductors of heat	They do not become hot.
7	**(c)** opaque	Wood and metal are opaque.
8	**(b)** wooden block	Wood is less dense than water.
9	**(a)** wool	Wool comes from sheep.
10	**(c)** translucent	Frosted glass is translucent.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The substance an object is made of (1); Wood and iron (1).	1+1
12	They are poor conductors of heat (1); So hands do not get burnt (1).	1+1
13	It is transparent (1); We can see through it (1).	1+1
14	Transparent: light passes through (1); Opaque: light is blocked (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Clear glass: transparent (1); Butter paper: translucent (1); Wooden door: opaque (1).	1+1+1
16	Cloth is a poor conductor (1); Prevents burns (1); Heat is not transferred fast (1).	1+1+1
17	Hard materials are not easily scratched (1); Used in tools, buildings (1); Soft materials used for cushions (1).	1+1+1
	OR: Take a glass of water (1); Add a spoon of the substance and stir (1); If it disappears, it is soluble (1).	1+1+1
18	Heat one end of a rod (1); Place a wax-held pin on the other end (1); If pin falls quickly, it is a good conductor (1).	1+1+1
	OR: Metal conducts heat (1); Heat travels along the spoon (1); Plastic spoons stay cool (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Appearance: lustre of gold (1); Hardness: diamond (1); Solubility: sugar (1); Transparency: glass (1); Floating: cork, sinking: stone (1).	1+1+1+1+1
	OR: Place objects in water (1); Wood, cork float (1); Stone, nails sink (1); Float if lighter than water (1); Record observation (1).	1+1+1+1+1
20	Collect objects (1); Hold each in front of a torch (1); Observe the light passing (1); Group as transparent, translucent, opaque (1); Record in a table (1).	1+1+1+1+1
	OR: By appearance (1); By hardness (1); By solubility (1); By conduction (1); Helps in selection (1).	1+1+1+1+1

Mistakes to avoid:

- Calling frosted glass or tracing paper transparent; they are **translucent**.
- Saying that insoluble means the substance cannot mix with anything; it means it does not dissolve in **water**.
- Writing that all metals are hard; sodium is soft, and only most metals are hard.
- Confusing **opaque** with translucent: opaque blocks all light.
- Thinking oil is soluble because it spreads on water; it floats as a separate layer.
- Assuming all shiny objects are metals; lustre alone is not proof, check conduction too.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** iron nail	Iron is denser.
2	**(d)** plastic	Plastic is synthetic.
3	**(d)** clear glass	Clear glass allows light through.
4	**(b)** soft	Chalk is soft.
5	**(a)** natural fibres	They come from plants and animals.
6	**(c)** tracing paper	Light passes partly.
7	**(a)** diamond	Diamond is the hardest natural substance.
8	**(c)** transparent	We can see through it.
9	**(d)** brick	Brick blocks light.
10	**(d)** cotton wool	Cotton is soft.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Shiny surface of a material (1); Gold and silver (1).	1+1
12	It does not allow water to pass (1); It is waterproof (1).	1+1
13	Transparent: glass, air; translucent: tracing paper (1); Opaque: wood, brick (1).	1+1
14	Floating materials are less dense than water (1); Sinking ones are denser (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It makes study easy (1); We can choose suitable materials (1); We find uses for each (1).	1+1+1
16	Plastic is an insulator (1); Does not pass electricity (1); Safe for hands (1).	1+1+1
17	They have lustre (1); Gold and silver are shiny (1); They do not rust easily (1).	1+1+1
	OR: Metals conduct heat (1); They are strong (1); Wood burns and is a poor conductor (1).	1+1+1
18	Natural: from plants/animals/earth (1); Man-made: prepared in factories (1); Cotton vs nylon (1).	1+1+1
	OR: Chair needs strength (1); Window needs transparency (1); Properties decide use (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Spoon: metal (1); Raincoat: plastic (1); Window: glass (1); Tyre: rubber (1); Properties decide use (1).	1+1+1+1+1
	OR: Bricks: strong (1); Cement: binds (1); Wood: doors (1); Glass: windows (1); Iron: grills (1).	1+1+1+1+1
20	Take 5 glasses of water (1); Add sugar, salt, sand, chalk, soda (1); Stir each (1); Sugar, salt, soda dissolve (1); Sand and chalk do not (1).	1+1+1+1+1
	OR: Cotton, wool, wood: natural (1); Plastic, nylon: man-made (1); Natural from plants or animals (1); Man-made in factories (1); Both useful (1).	1+1+1+1+1

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- Writing that all metals are hard; sodium is soft, and only most metals are hard.
- Confusing **opaque** with translucent: opaque blocks all light.
- Thinking oil is soluble because it spreads on water; it floats as a separate layer.
- Assuming all shiny objects are metals; lustre alone is not proof, check conduction too.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** control the light entering a room	They reduce glare and give privacy.
2	**(c)** lustre	Metals look shiny.
3	**(c)** rough	Sandpaper is rough.
4	**(c)** waterproof	They do not let water pass through.
5	**(d)** soluble	Salt is soluble.
6	**(d)** glass sheet	Glass is smooth.
7	**(a)** copper	Metals can be drawn into wires.
8	**(b)** insoluble	Sand is insoluble.
9	**(c)** conductors of heat	Metals conduct heat.
10	**(d)** gold	Gold shines.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Copper conducts electricity well (1); It can be drawn into wires (1).	1+1
12	Soluble dissolves in water (1); Insoluble does not dissolve (1).	1+1
13	Natural: cotton, wood (1); Man-made: plastic, nylon (1).	1+1
14	It gives a good grip (1); It prevents slipping (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lustre (1); Hardness (1); Good conduction of heat (1).	1+1+1
16	Glass is waterproof (1); Does not react (1); Strong and transparent (1).	1+1+1
17	Use a magnet (1); Nails stick (1); Wood remains (1).	1+1+1
	OR: Float: cork, wood, thermocol (1); Sink: iron nail, stone, coin (1); Depends on density (1).	1+1+1
18	Sugar dissolves (1); Sand settles (1); Sugar is soluble; sand is insoluble (1).	1+1+1
	OR: Steel is waterproof and strong (1); Paper gets wet (1); Steel is durable (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conductors pass heat easily (1); Examples: iron, copper (1); Insulators do not pass heat (1); Wood, plastic (1); Used on handles (1).	1+1+1+1+1
	OR: Rough: sandpaper (1); Smooth: glass (1); Rough surface gives grip (1); Smooth reduces friction (1); Used in floors and tools (1).	1+1+1+1+1
20	Lustre (1); Hard and strong (1); Can be beaten into sheets (1); Drawn into wires (1); Used in vessels, wires, bridges (1).	1+1+1+1+1
	OR: Collect the materials (1); Choose a property (1); Group objects (1); Record the groups (1); State the use of each (1).	1+1+1+1+1

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